

APDTIA4B-10G-W

OVERVIEW

Avalanche photodiode with a low-noise transimpedance amplifier with auto gain control coupled to an optical fiber and packaged into a hermetic case.

MAIN FEATURES

- High-sensitivity avalanche photodiode with limiting TIA
- Operation wavelength 1250 – 1650 nm
- Data rate: 10 Gbps
- Sensitivity: -29 dBm
- Package types: coaxial with or without bracket
- Low back reflection, return loss RL = 45 dB

ORDERING INFORMATION

APDTIA4B-10G-W-X-X-X-X-X-X

Optical matching

R30: back reflection -30 dB (SM1 and SM3 fiber)

LS: assembly with MM5 fiber, back reflection -15 dB

Case type

COAX: compact coaxial (low duty cycle pulse mode only)

COAXB: compact coaxial with a bracket

Pinout code

10: see more details on page 4

Fiber type

SM1: SM, G.657.A1, [Corning SMF-28 Ultra](#), furcation tubing Ø0.9 mm or **BSM1** Ø0.25mm

SM3: SM, G.657.B3, [Corning ClearCurve ZBL](#), furcation tubing Ø0.9 mm or **BSM3** Ø0.25mm

MM5: MM, [50/125. OM3](#), furcation tubing Ø0.9 mm

Other type on request

Connector type

FU: FC/UPC (SM1, SM3, MM5)

FA: FC/APC (SM1, SM3, MM5)

SU: SC/UPC (SM1)

SA: SC/APC (SM1)

N: no connector (scissors cut)

Other type: on request

Fiber length

0.5: 500+/-50 mm

1.0: 1000+/-100 mm

Other length on request

APDTIA4B-10G-W

ABSOLUTE MAXIMUM RATINGS

Parameter		Value	Unit	Conditions
TIA supply voltage	V _{CC}	-0.5 to 5.0	V	
APD voltage	V _{APD}	V _{BR}		
APD reverse current	I _R	2	mA	
Input optical power damage limit	P _{dam}	-4	dBm	
Operating temperature*	T _{op}	-40 - +85	°C	
Storage temperature	T _{stg}	-40 - +85	°C	
Soldering temperature	T _{sold}	260	°C	max 5 seconds

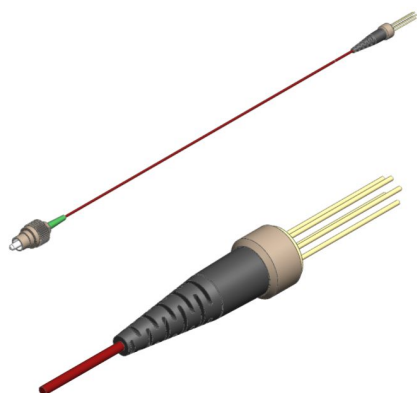
*Operating temperature is defined by the case temperature. It is necessary to ensure sufficient heat dissipation so that the module's maximum operating temperature is not exceeded. Operation at elevated temperatures reduces the lifetime of the module.

APDTIA4B-10G-W

ELECTRICAL-OPTICAL CHARACTERISTICS (T = 25 °C)

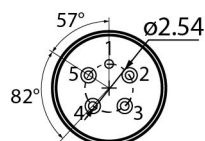
Parameter		MIN	TYP	MAX	Unit	Conditions
Operating wavelength	λ	1250		1650	nm	
Responsivity @ unity gain M = 1	R	0.85	0.95		A/W	$\lambda = 1550\text{nm}$, P = 1uW
Return loss	RL	25	30		dB	R30, SM1, SM3
Return loss	RL		15		dB	LS, MM5
Breakdown voltage	V _{BR}	21	27	33	V	I _d = 10 uA
Breakdown voltage temperature coefficient $\Delta V_{BR}/\Delta T$	δ		0.03		V/C	T = 25 C
Dark current	I _d		100	300	nA	V _R = 0.9 V _{BR}
TIA supply voltage	V _{cc}	3.0	3.3	3.46	V	
TIA supply current	I _{cc}		27	38	mA	
Data rate			10		Gbps	
Low-frequency cut-off	f _c		14	30	kHz	
Optical sensitivity	P _{min}		-29	-27.5	dBm	BER = 10 ⁻¹² , BR = 10.3125 Gbps, PRBS = 2 ³¹ -1, ER = 13 dB, $\lambda = 1550\text{ nm}$
Optical sensitivity	P _{min}		-25.5	-24	dBm	BER = 10 ⁻¹² , BR = 10.3125 Gbps, PRBS = 2 ³¹ -1, ER = 6 dB, $\lambda = 1550\text{ nm}$
Maximum output voltage	V _{out}	250	300	350	mVp-p	Differential
Transimpedance gain	Z _t	2.5	4.0		kOhm	
Output impedance	Z _{out}	40	50	60	Ohm	
Overload	P _{max}	-5			dBm	BER = 10 ⁻¹² , BR = 10.3125 Gbps, PRBS = 2 ³¹ -1, ER = 6 dB, $\lambda = 1550\text{ nm}$

APDTIA4B-10G-W



COAX

BACK VIEW



PINOUT

#10

1. Gnd
2. Dout
3. Vcc
4. Vapd
5. Dout

Download more
information



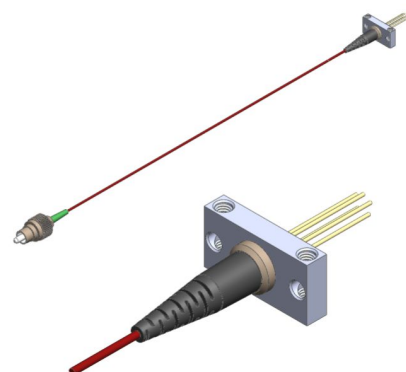
Drawing



3D model

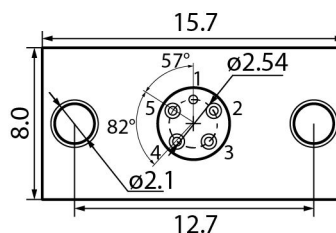


Application Notes



COAXB

BACK VIEW



PINOUT

#10

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Drawing



3D model



Application Notes

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Characteristics, data, materials and structures specified in this datasheet are subject to change without notice. Please refer to the latest specification before use of the products.

Safety and handling cautions

1. Avoid smashing and burning of the module. Avoid storing and using the module in conditions where water, organic solvents or aggressive acids or bases may contact the module or where there is a possibility of exposure to corrosive gases, explosive gases, dust, salinity or other harsh conditions. The module should be disposed as special industrial waste.
2. Exceeding absolute maximum ratings even for a short time can cause permanent damage of the module.
3. The module is sensitive to and can be broken by ESD (static electricity).

Conflict Minerals Policy Statement

LD4B achieves business objectives and customer needs with social responsibility. We do not support or contribute to the violence and human rights violations associated with the mining of conflict minerals coming from Conflict Regions according to US "Dodd-Frank Act". When possible, our suppliers' conflict mineral statements are reviewed. We do not directly purchase Conflict Minerals from any source and do not knowingly procure any parts and products containing Conflict Minerals from Conflict Regions.

RoHS Compliance Statement

Restriction of Hazardous Substances (RoHS) directive (Directive 2011/65/EC amended with Directive (EU) 2015/863) is the directive aimed at reducing the harmful environmental impact of waste electrical equipment by restricting the use of known dangerous substances. Based on information received from our supply sources, LD4B hereby states that the banned substances listed in the RoHS directive are not found in the parts and materials used above the threshold level listed other than exceptions approved by the European Commission.

REACH Compliance Statement

Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH) is a European Union regulation 1907/2006/EC that addresses the production and use of chemical substances, and their potential impacts on human health and the environment. Based on information received from our supply sources, LD4B hereby states compliance of the parts and materials used in manufacturing to REACH regulation. LD4B does not manufacture or import any substances or preparations as defined under REACH.